Entity and Relationship Analysis Report

E-Commerce Order Management Database System — Week 2 Deliverable

1. Introduction

This document analyzes the relationships between the nine core entities of the E-Commerce Order Management Database System, as identified in the companion Entity Analysis Report, and specifies the cardinality of each relationship. It concludes with a consolidated Entity-Relationship (ER) diagram that will serve as the direct input to the physical database design and SQL implementation in the following weeks.

2. Relationship Analysis

The following relationships were identified by tracing each foreign key back to the primary key it references, and by analyzing the business processes documented in the Week 1 SRS.

Relationship	Description
Category — Product	A Category classifies many Products; each Product belongs to exactly one Category. Implemented via Product.Category_ID → Category.Category_ID.
Customer — Order	A Customer can place many Orders; each Order is placed by exactly one Customer. Implemented via Order.Customer_ID → Customer.Customer_ID.
Customer — Review	A Customer can write many Reviews; each Review is written by exactly one Customer. Implemented via Review.Customer_ID → Customer.Customer_ID.
Product — Review	A Product can receive many Reviews; each Review refers to exactly one Product. Implemented via Review.Product_ID → Product.Product_ID.
Order — Order Details	An Order can contain many Order Details lines; each Order Details line belongs to exactly one Order. Implemented via Order Details.Order_ID → Order.Order_ID.
Product — Order Details	A Product can appear in many Order Details lines across different orders; each Order Details line refers to exactly one Product. Implemented via Order Details.Product_ID → Product.Product_ID. Together with the relationship above, Order Details resolves the underlying many-to-many relationship between Order and Product.
Order — Payment	Each Order has exactly one Payment record, and each Payment record belongs to exactly one Order. Implemented via Payment.Order_ID → Order.Order_ID, with Order_ID marked UNIQUE in Payment.
Order — Shipment	Each Order has exactly one Shipment record, and each Shipment record belongs to exactly one Order. Implemented via Shipment.Order_ID → Order.Order_ID, with Order_ID marked UNIQUE in Shipment.
Supplier	No foreign key relationship can be established between Supplier and any other entity using the fixed Week 1 attribute list, since Product does not carry a Supplier_ID attribute. Supplier is therefore analyzed as a standalone master entity in the current schema (see Section 5 of the Entity Analysis Report).

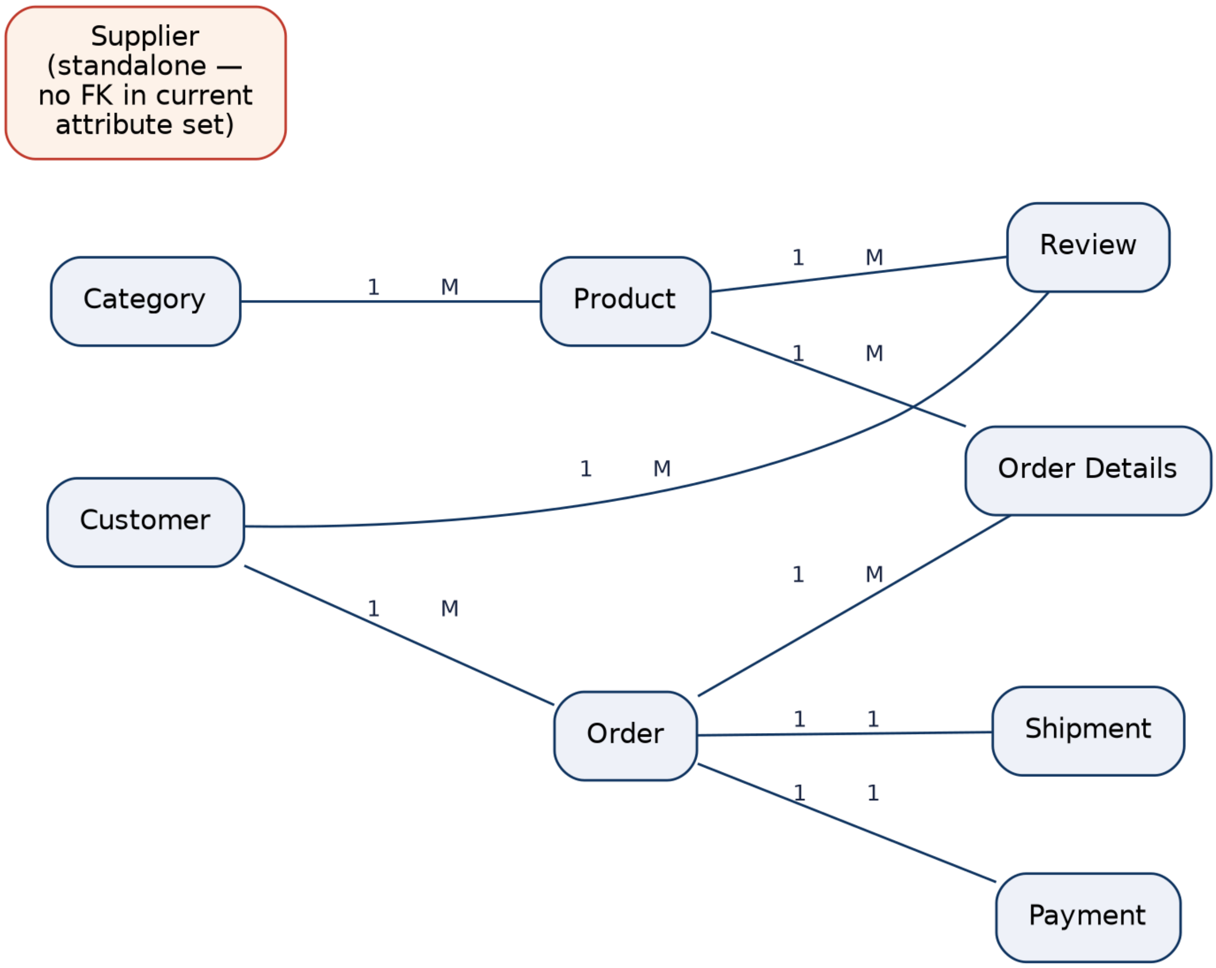
3. Cardinality Analysis

The table below states the cardinality (One-to-One, One-to-Many, or Many-to-Many) of every relationship identified in Section 2, together with the participation (mandatory/optional) of each side.

Entity A	Entity B	Cardinality	Participation Rule
Category	Product	One-to-Many (1:M)	A Category may classify zero or many Products; a Product must belong to exactly one Category.
Customer	Order	One-to-Many (1:M)	A Customer may place zero or many Orders; an Order must belong to exactly one Customer.
Customer	Review	One-to-Many (1:M)	A Customer may write zero or many Reviews; a Review must belong to exactly one Customer.
Product	Review	One-to-Many (1:M)	A Product may receive zero or many Reviews; a Review must refer to exactly one Product.
Order	Order Details	One-to-Many (1:M)	An Order must have one or many Order Details lines; each line must belong to exactly one Order.
Product	Order Details	One-to-Many (1:M)	A Product may appear in zero or many Order Details lines; each line must refer to exactly one Product.
Order	Product	Many-to-Many (M:N)	Resolved through Order Details: an Order may include many Products, and a Product may appear in many Orders.
Order	Payment	One-to-One (1:1)	An Order must have exactly one Payment record, and a Payment record must belong to exactly one Order.
Order	Shipment	One-to-One (1:1)	An Order must have exactly one Shipment record, and a Shipment record must belong to exactly one Order.

4. Entity-Relationship Diagram

The diagram below consolidates all identified entities and relationships. Boxes represent entities; connecting lines are labelled with their cardinality (1 = one, M = many). Supplier is shown separately since the fixed attribute list does not currently provide a foreign key linking it to any other entity.



5. Summary

Nine entities were analyzed, yielding eight enforceable relationships: six one-to-many relationships, two one-to-one relationships, and one many-to-many relationship (between Order and Product) that is resolved through the Order Details associative entity. Supplier remains an unlinked master entity under the current fixed attribute list. This analysis, together with the Entity Analysis Report, provides the complete data model required to proceed to schema normalization and SQL implementation in the subsequent project weeks.